

Name: _____ Date: _____

Period: _____

Primary Objective**By the end of this lesson, I will be able to...**

- solve multistep linear equations and inequalities in one variable, *naming the property* behind every step;
- decide whether a linear equation has *one, no, or infinitely many* solutions;
- graph an inequality's solution set on a number line and rearrange a *literal* equation for a specified variable;
- write an equation or inequality from a situation and interpret the answer back in that situation;
- solve a quadratic by square roots, by factoring with the *zero product property*, and by the *quadratic formula*;
- use the *discriminant* to say how many real solutions exist before solving — and match it to the graph.

Vocabulary & Key Concepts

Fill in each term as we build it together.

Equation (vs. expression) / solution set:

Two expressions joined by $=$. An expression is a recipe and is *evaluated*; an equation asks a question and is *solved*. Its solution set is the set of all values that make it true......

Properties of equality:

Add, subtract, multiply, or divide both sides by the same quantity (never divide by 0). They are the only moves that produce an *equivalent* equation — same solution set, easier wording......

Identity vs. contradiction:

An identity is true for every value (both sides are the same expression), so the solution set is all real numbers. A contradiction is true for no value, so the solution set is empty......

Literal equation:

An equation containing several letters, solved for one specified variable — e.g. $A = \frac{1}{2}bh$ solved for h gives $h = \frac{2A}{b}$

Properties of inequality:

Identical to the properties of equality with one exception: multiplying or dividing both sides by a *negative* number reverses the inequality symbol......

Zero product property:

If $mn = 0$ then $m = 0$ or $n = 0$. It is what turns a factorization into solutions — and it works only against zero......

Discriminant:

The quantity $b^2 - 4ac$ under the radical in the quadratic formula. Positive gives two real solutions, zero gives one, negative gives none......

Hook**Solve all three.** Nothing else — just solve them.

(a) $3(x - 2) = x - 6$

$3(x - 2) = x - 6$

$3x - 6 = x - 6$

$2x - 6 = -6$

$2x = 0$

$x = 0$

$$(b) \quad 3(x - 2) = 3x - 6$$

$$3(x - 2) = 3x - 6$$

$$3x - 6 = 3x - 6$$

$$-6 = -6$$

$$(c) \quad 3(x - 2) = 3x - 5$$

$$3(x - 2) = 3x - 5$$

$$3x - 6 = 3x - 5$$

$$-6 = -5$$

Two of those did something you were not expecting: the variable **disappeared**. That is not a mistake — it is the answer arriving in a different costume.

- In (b) you were left with a statement that is **true**, so how many values of x work? **all of them**
- In (c) you were left with a statement that is **false**, so how many values of x work? **none**

Now the same question in another family. How many real solutions does each have? $x^2 = 9$: **2**
 $x^2 = 0$: **1** $x^2 = -9$: **0**

The idea the whole lesson hangs on: an expression was a *recipe*; an equation is a **question**. The answer is a *solution set*, and it is allowed to hold one number, no numbers, or every number. By the end of today you will know the count **before** you solve.

1. Linear Equations — and How Many Answers They Have

An **equation** joins two expressions with $=$. Unlike an expression, it asks a **question**, and a **solution** is a value that makes it **true**.

Solve, and name every move. The *properties of equality* are the only moves that keep the solution set the same — which is exactly why they are allowed and nothing else is.

Line	Property that makes it legal
$5x + 3(x - 2) = 2(x + 9)$	<i>given</i>
$5x + 3x - 6 = 2x + 18$	distributive property
$8x - 6 = 2x + 18$	<i>combine like terms</i>
$6x - 6 = 18$	subtraction property of equality (subtract $2x$)
$6x = 24$	addition property of equality (add 6)
$x = 4$	division property of equality (divide by 6)

Verify it. Substitute your answer into the *original* equation — not into a line partway down, where a mistake would already be baked in.

Left side: $5(4) + 3(4 - 2) = 20 + 6 = 26$ Right side: $2(4 + 9) = 26$ Equal? **yes**

Checking is not politeness. It is the **definition** of the word “solution.”

Back to the Hook: three possible answers. Fill in what the *last line* tells you.

Equation	Last line	Solution set	Name
$3(x - 2) = x - 6$	$x = 0$	one solution: $x = 0$	—
$3(x - 2) = 3x - 6$	$-6 = -6$	all real numbers	identity
$3(x - 2) = 3x - 5$	$-6 = -5$	no solution (empty)	contradiction

If the variable survives → **exactly one** solution. If it vanishes leaving a **true** statement → **all real numbers**.
 If it vanishes leaving a **false** statement → **no solution**.

Write one from a situation. A repair shop charges a \$40 flat fee plus \$15 per hour of labor. A bill came to \$130.

Equation: $40 + 15h = 130$ Solve: $h = 6$

Now answer the *question that was asked*, in a sentence with units: **The repair took 6 hours of labor.** (“ $x = 6$ ” is not an answer to the question that was asked — the question was about hours.).....

2. Literal Equations — Same Moves, Scarier Alphabet

A **literal equation** has several letters. “Solve for h ” means: get h **by itself on one side**, using the same properties of equality as always.

Formula	Solve for	Result
$A = \frac{1}{2}bh$ (area of a triangle)	h	$h = \frac{2A}{b}$
$P = 2\ell + 2w$ (perimeter)	w	$w = \frac{P - 2\ell}{2}$
$y = mx + b$	x	$x = \frac{y - b}{m}$
$C = \frac{5}{9}(F - 32)$	F	$F = \frac{9}{5}C + 32$

The one real trap. In $P = 2\ell + 2w$, are 2ℓ and $2w$ like terms? **No** So you may *not* write $P = 4\ell w$. And when you divide, you must divide **every** term, not just the one you were looking at.

Coming attraction: in Unit 2 you will meet $y = mx + b$ rearranged three different ways and hear them called “forms.” They are not three facts — they are one equation, solved for different things.

3. Linear Inequalities — and Why the Symbol Flips

Everything about solving is the same ... except for one move. Let’s find it rather than announce it.

Start with a statement that is definitely true: $2 < 6$.

Multiply *both sides* by -1 : $-2 > -6$ (write in the symbol that is actually true)

What happened to the order? **It reversed — the bigger number became the smaller one.**

On the number line, multiplying by a negative **reflects** everything across zero — and a reflection **reverses** the order. So the rule is not a convention somebody chose:

Multiplying or dividing both sides by a negative number reverses the inequality symbol. (Adding or subtracting a negative does *not* — only multiplying and dividing.)

Solve and graph. Shade the solution set on the number line. Endpoint circles: *open* $\circ$ for $<$ and $>$, *closed* $\bullet$ for $\leq$ and $\geq$.

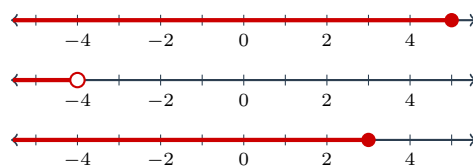
Solve

(a) $4x - 9 \leq 11$ $x \leq 5$

(b) $-3x + 7 > 19$ $x < -4$

(c) $2(x + 4) \geq 6x - 4$ $x \leq 3$

Graph the solution set



Which one made you flip the symbol, and at exactly which step? **Item (b), at the last step: dividing both sides of $-3x > 12$ by -3 . Note that (c) does *not* flip if you collect the x terms on the right ($12 \geq 4x$) — both routes are legal and both end at $x \leq 3$.** In (c) you may end with $3 \geq x$. Rewrite it as $x \leq 3$ before you graph — reading the shading off a backwards inequality is how the picture ends up on the wrong side.

Write one from a situation. Concert tickets cost \$12 each and you can spend **at most** \$75.

“At most” translates to the symbol $\leq$. Inequality: $12t \leq 75$ Solve: $t \leq 6.25$

The algebra says $t \leq 6.25$. How many tickets can you actually buy, and why is the answer not 6.25? **At most**

6 tickets. Tickets come in whole numbers, so the context restricts t to non-negative integers; the algebra's answer has to be interpreted, not just reported.

4. Quadratic Equations — Three Methods and a Head Count

A **quadratic equation** can be written $ax^2 + bx + c = 0$ with $a \neq 0$. Which method you use depends on what the equation *looks like*.

Method 1 — Square roots (use when there is no x term). Isolate the square, then take the root of both sides. **Both** roots: the $\pm$ is not optional.

$$(a) \quad x^2 = 49, \quad x = \pm 7 \quad (b) \quad 2x^2 - 50 = 0, \quad x = \pm 5 \quad (c) \quad (x - 1)^2 = 16, \quad x = 5 \text{ or } -3$$

Writing only $x = 7$ in (a) does not lose a step — it loses **a solution**.

Method 2 — Factor, then the zero product property (use when it factors).

Zero product property: if $mn = 0$ then $m = 0$ or $n = 0$.
Step 0, every single time: make one side equal **0** first.

Equation	Set to zero, then factor completely	Solutions
$x^2 + 2x - 15 = 0$	$(x + 5)(x - 3) = 0$	$x = -5, 3$
$x^2 - 3x = 10$	$x^2 - 3x - 10 = 0$, so $(x - 5)(x + 2) = 0$	$x = 5, -2$
$2x^2 - 18 = 0$	$2(x + 3)(x - 3) = 0$	$x = \pm 3$

Why “completely” matters more now. In Lesson 1.1, stopping at $2(x^2 - 9)$ cost you a step. Here it costs you **a solution**.

And why the zero is not negotiable: from $(x - 2)(x + 5) = 8$, may you conclude $x - 2 = 8$? **No** Why not? **Eight can be written as a product of two nonzero numbers in infinitely many ways** ($8 \cdot 1$, $4 \cdot 2$, $16 \cdot \frac{1}{2}$, ...), so knowing the product is 8 forces nothing about either factor. **Zero is the only number that cannot be built from two nonzero factors — which is the entire reason the property exists.**

Method 3 — The quadratic formula (use when nothing else works — it always works).

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solve $x^2 - 4x - 7 = 0$. $a = 1$ $b = -4$ $c = -7$

Substitute and simplify:

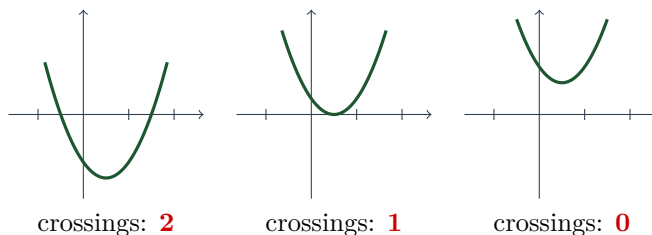
$$\begin{aligned} x &= \frac{4 \pm \sqrt{(-4)^2 - 4(1)(-7)}}{2(1)} \\ &= \frac{4 \pm \sqrt{16 + 28}}{2} \\ &= \frac{4 \pm \sqrt{44}}{2} \\ &= \frac{4 \pm 2\sqrt{11}}{2} \\ &= 2 \pm \sqrt{11} \end{aligned}$$

— these are **irrational** solutions, and simplifying that radical is Lesson 1.1 doing its job.

The head count: the discriminant. The piece under the radical, $b^2 - 4ac$, decides everything *before* you solve. Same a and b below — only c changes.

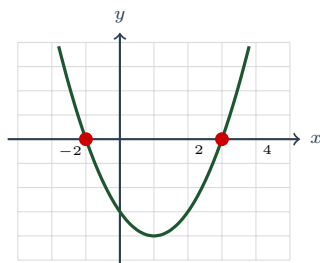
Equation	$b^2 - 4ac$	Positive / zero / negative?	# real solutions
$x^2 - 2x - 3 = 0$	16	positive	two
$x^2 - 2x + 1 = 0$	0	zero	one
$x^2 - 2x + 3 = 0$	-8	negative	none

The same three facts, drawn. Below are the graphs of those three, in order. Label each one with the number of times it crosses the x -axis.



The discriminant and the picture are the **same** fact told twice — a real solution *is* an x -intercept.

Read the graph, then confirm with algebra. This is $y = x^2 - 2x - 3$.



Read the x -intercepts off the graph: $x = -1$ and $x = 3$

Now confirm by factoring $x^2 - 2x - 3 = 0$: $(x - 3)(x + 1) = 0$

Do they agree? **yes** Finish the sentence: solving $x^2 - 2x - 3 = 0$ is the same task as finding the **x -intercepts (zeros)** of the graph.

Guided Practice

1. Solve and check: $7(x - 2) = 3x + 6$.

$$7(x - 2) = 3x + 6$$

$$7x - 14 = 3x + 6$$

$$4x - 14 = 6$$

$$4x = 20$$

$$x = 5$$

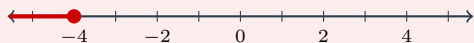
Check: $7(3) = 21 = 3(5) + 6$

2. How many solutions does $2(3x + 4) = 6x + 8$ have? **infinitely many** How do you know? **both sides are $6x + 8$ — an identity**
3. Solve $-5x + 3 \geq 23$ and graph the solution set.

$$-5x + 3 \geq 23$$

$$-5x \geq 20$$

$$x \leq -4$$



4. Solve for r : $C = 2\pi r$. $r = \frac{C}{2\pi}$

5. Without solving, how many real solutions does $x^2 + 4x + 9 = 0$ have? **none** Justify with a number:
 $b^2 - 4ac = 16 - 36 = -20 < 0$
6. Solve by factoring: $x^2 + x = 12$.

$$\begin{aligned}x^2 + x &= 12 \\x^2 + x - 12 &= 0 \\(x + 4)(x - 3) &= 0 \\x &= -4 \text{ or } x = 3\end{aligned}$$

Verify one solution: $3^2 + 3 = 9 + 3 = 12$ **true**

Teacher Note

The Hook is the lesson. Do not rescue students from the vanishing variable in (b) and (c). The entire day depends on them experiencing “I got $-6 = -5$, so I did it wrong” and being told *no, you finished*. Write “one / none / all” on the board next to “two / one / none” and leave both up for the whole period.

Section 1’s property column is graded work, not decoration. **A.EI.1b** says students solve *by applying the properties*, so accepting “I moved it over” quietly de-implements the standard. This is the third lesson in a row asking for the same column; by 1.3 it should be automatic.

Section 3: derive the flip, never assert it. The $2 < 6 / -2 > -6$ demonstration takes ninety seconds and it is the difference between a rule students forget under pressure and a fact they can rebuild. The reflection language pays off again in Unit 2, where $y = -x$ is the same reflection acting on a whole graph.

Section 4, the two errors that cost real points. (i) Not setting the equation to zero before factoring — $x^2 - 3x = 10$ becoming $x(x - 3) = 10$ and then “ $x = 10$ or $x - 3 = 10$ ” is the single most common quadratic error on the SOL. (ii) Dropping the negative root in Method 1. Both are diagnosable from the notes table at a glance while circulating.

The discriminant before the graphs, not after. Have the room compute all three discriminants first, predict the crossings out loud, *then* look at the three parabolas. If the figures are shown first, students read the answer off the picture and the predictive point — the whole content of **A.EI.3b** — is lost.

Where this goes next. Lesson 1.3 defines the *zeros* of a function as the solutions of $f(x) = 0$. Every technique in Section 4 is the machinery for finding x -intercepts, and the inequality work in Section 3 is how domain restrictions get written. Say that explicitly at the close so 1.3 opens as a continuation rather than a new topic.